

In Exercises 23-26, describe the possible echelon forms of the matrix. Use the notation of Example 1 in Section 1.2.

23) A is a 3×3 matrix with linearly independent columns.

Solution: Since A has linearly independent columns, the matrix equation $Ax=0$ has ^{only} the unique trivial solution. By Theorem 3, the matrix eqn. has the same solution set as the system of equations corresponding to the augmented matrix $[A \ 0]$. By the row reduction algorithm, this matrix can be converted to rref and by Theorem 1, the rref form is unique. By Theorem 2, the matrix A has no free variables. $\therefore$ Each column of matrix A must have a pivot, by defn. By defn. of echelon form and pivot, each row's pivot must be in a column to the right of the columns of the pivots of all rows above it.

From these facts, it's easy to conclude that the pivot entries are in Row 1 Column 1, Row 2 Column 2 and Row 3 Column 3.

$\therefore$ In echelon form, Row 1 Column 1 = Row 2 Column 2 = Row 3 Column 3 =

$\square$ By defn. of echelon form, for all ~~rows~~ ^{columns} having leading elements, all positions below it must contain 0.

$\therefore$ Row 2 Column 1 = Row 3 Column 1 = 0
Row 3 Column 2 = 0

There are no constraints on the positions of Row 1 Column 2, Row 1 Column 3 and Row 2 Column 3

$\therefore$ Row 1 Column 2 = Row 1 Column 3 = Row 2 Column 3 = *

The final echelon form of matrix A is
$$\begin{bmatrix} \square & * & * \\ 0 & \square & * \\ 0 & 0 & \square \end{bmatrix}$$

24) A is a 2×2 matrix with linearly dependent columns.

Solution: Let the columns of A be a_1 and a_2 . Since the columns are linearly dependent, by defn, $\exists c_1, c_2 \in \mathbb{R}$ where not both c_1 and c_2 are 0 together, and $c_1 a_1 + c_2 a_2 = 0$. By defn. of matrix equation, $[a_1 \ a_2] \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.